Indoor maps for wayfinding applications & Simple Indoor Tagging in OpenStreetMap

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I. Introduction

Society has come to heavily rely on modern navigation applications. It is incredibly useful for electronics to be able to identify a user's current location and instruct them to a desired location. This is only made possible by the level of detail in current maps. The data, and surrounding software, that is used in maps (geographic information systems) has recently grown at a rapid rate and is used for many applications and purposes.

The future of mapping is faced with mapping indoor spaces. There is a vast need for indoor mapping and there are many applications that could benefit from adequate detail. For example, airport travelers will be able to find a list of the nearby shops and restaurants, as well as how to get there.

People with disabilities, such as mobility impairments and vision impairments, will particularly benefit from indoor mapping. A person in a wheelchair may need to know the closest elevator or ramp. A blind person may need to know where a certain room is located and will need verbal instructions, without the use of a screen. There are countless uses of indoor maps that make it possible to improve the lives of people around the world.

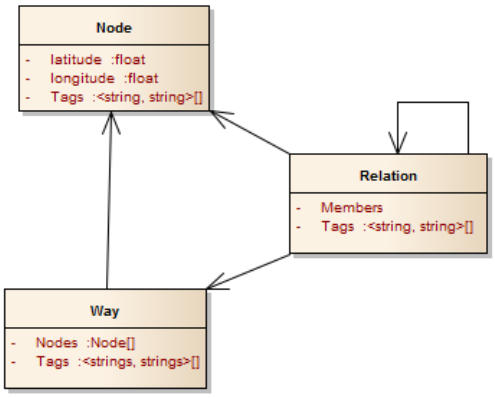
Indoor maps that are widely available, and easily obtainable, promote growth of applications that are inclusive of disability accommodations. This paper will discuss the potential of indoor mapping in OpenStreetMap, a geographic database, and will demonstrate the status of indoor mapping in several cities and in USA.

II. Previous Work

A. OpenStreetMap

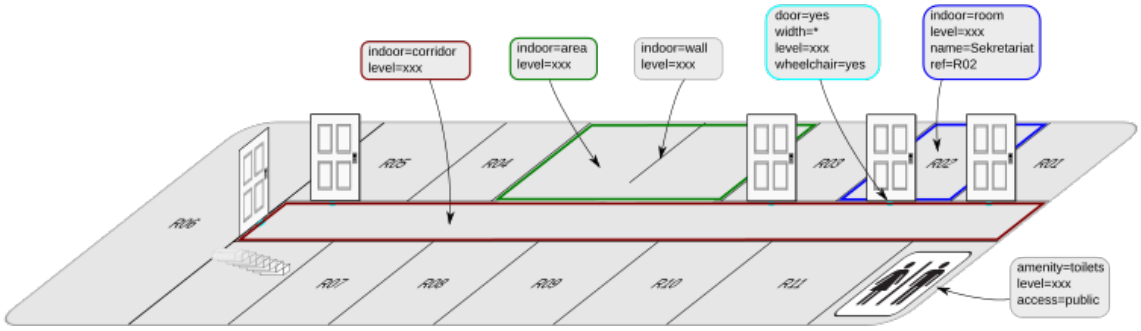
The navigation applications that have become most familiar to the common user are Google Maps, Apple Maps, and those alike. Most of these companies have begun growing databases of indoor spaces, however, they have proprietary data that prevents further analysis and development [11]. Open-source maps are more favorable for the future of indoor mapping, especially in applications for people with disabilities, because they allow the spread of the information freely. Accommodations for disabilities have always lagged behind the rest of industry since there is often less of a financial market for accommodating disabilities.

OpenStreetMap (OSM) is an open-source geographic database and is a promising option for providing accessible indoor mapping information for people with disabilities. It also has the advantage of being created by crowdsourcing, allowing the input of many people around the world collaborating with different perspectives and objectives. This open-source, crowdsourced approach allows the flexibility to push forward the future of indoor accessibility.



[Figure 1. OpenStreetMap data model elements.
https://docs.itinero.tech/docs/osmsharp/osm.html]

OpenStreetMap has a very simple data model. It consists of 3 data structure elements: nodes, ways, and relations. A node is a single point on the map, and a way is a list of nodes, creating lines. A relation is a list of nodes, ways, and tags. Additionally, each of these 3 elements can be associated with



[Figure 2. OSM SIT tagging elements example.
https://wiki.openstreetmap.org/wiki/SimpleIndoorTagging]

any number of key-value pairs, called tags (shown in "Fig. 1"). Tags are responsible for identifying all the features on a map (labeling what each element represents). Although the use of tags is generally unconstrained, there are conventional use of tags in OSM.

This structure is key to the success of the map and provides a great environment for indoor accessibility information.

B. Indoor mapping in OpenStreetMap

Expressing indoor spaces in a database is a complicated endeavor. There are many of approaches and the science is constantly evolving. Within OSM, there are competing methodologies, however, the most prominent is a tagging schema called Simple Indoor Tagging (SIT) [12]. SIT tagging is a simple approach, much like the OSM data model. There are 5 elements that make up the SIT model: room, area, walls, corridor, and level "Fig. 2". It is also suggested that the minimum tagging requirement of each building have the lowest level (min_level) and highest level (max_level) [12].

C. Wayfinding Applications

Wayfinding applications must communicate directions to the user. The application must know both the current location and how to get to the destination. To do this, applications must utilize special information from geospatial databases such as Google Maps, Apple Maps, and OSM.

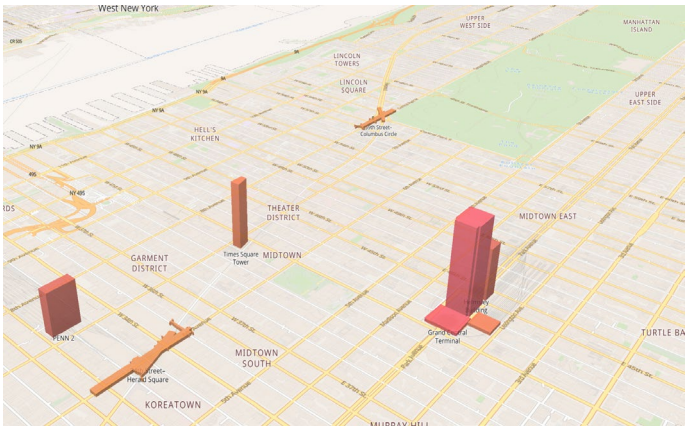
Some applications are made to accommodate wheelchair accessibility specifically, and require special information from a database, such as ramp and elevator locations. Wheelmap is an application created in Berlin, Germany which uses the OSM database and is also entirely crowd sourced [5]. Wheelmap identifies places (outdoor or indoor) as one of 3 categories: wheelchair accessible, not wheelchair accessible, or partially wheelchair accessible [15]. OpenSidewalks is an application created at the University of Washington which uses computer vision and neural networks to map outdoor sidewalks [8]. These applications target the needs of wheelchair users, specifically, because there is such extensive variety of disabilities and therefore a wide range of accommodations that need to be addressed.

Vision impairments often require more attention to detail when creating geospatial information systems [6]. Wayfinding applications rely on technology to identify its current location such as GPS. However, GPS most often lacks the precision required to identify specifically which room a person is standing within a building. Moreover, buildings often impede the GPS signals causing a further decrease in accuracy. To overcome this, researchers have investigated Bluetooth beacon technology, which has shown promising results. NavCog is an application created at Carnegie Mellon that is designed specifically for blind, or visually impaired, navigation by providing auditory guidance [7]. To achieve higher position accuracy, many Bluetooth beacons must be strategically placed throughout a building. A map database must include the location of these beacons as well as precise information describing the floor plans.

CityGuide takes indoor auditory guidance further by providing seamless integration between outdoor and indoor settings, which requires multiple technologies [3]. CityGuide was created by researchers at Wichita State University alongside another project, IBeaconMap, which uses computer vision to collect data from floor plans [2]. IBeaconMap uses the floor plan information to determine appropriate beacon locations, while also generating data to map the indoor spaces with adequate information for wayfinding applications.

AccessibleMaps is an impressive project that is based in Vienna, Austria [10]. AccessibleMaps has made substantial progress in the automation of indoor mapping and enriching the data with accessibility features [4]. This extends SIT tagging in OSM and focuses on the wide variety of disability needs. AccessibleMaps aims to provide highly customizable data for wayfinding applications, whereas each user's experience can be individualized according to their specific disability and accommodation needs.

[Figure 3. IndoorEqual level selector view of
Grand Central Terminal, New York New York, USA
https://indoorequal.org/ *Accessed Aug. 08, 2020*]



[Figure 4. OpenIndoor 3D view of
Grand Central Terminal, New York New York, USA
https://app.openindoor.io/ *Accessed Aug. 08, 2020*]

Creating viewable maps from OSM data is also important for navigation applications. Several projects are working on ways to render and view indoor maps, including accessibility data [19-21]. IndoorEqual is a viewer with a level selector feature shown in Fig. 3. OpenIndoor provides a 3D view of indoor data in OSM "Fig. 4".

III. Methods

Although much progress has been made thanks to the efforts of researchers such as those previously mentioned, there exists a dramatic shortage in data, making indoor wayfinding impractical on a large scale, especially for disability accommodations. This paper will demonstrate the amount of SIT tagging available on OSM.

To access OSM data, an API must be utilized such as Overpass [13]. This is a read-only API which is suitable for analyzing the current contents of OSM. Overpass operates with a query language that is capable of returning several formats of OSM data and is used in the backend of many applications [14]. Overpass is available for LINUX operating systems and can return millions of data elements within minutes. Since the data needed for this paper will not be used to create a map, Overpass Turbo is used for data collection which cannot return as much data elements as Overpass. Overpass Turbo is a web-based frontend application that is suitable for demonstrating the state of indoor mapping in OSM [9].

Information is collected for various cities in USA using Overpass Turbo. The number of elements in a city that contain the tag "building" is collected along with the key-value pair "indoor=yes". SIT tagging information is collected in each city which provides a better understanding of the status of indoor mapping in OSM. The number of elements with any of the 5 SIT elements is also collected.

Minimal SIT tagging is considered any building that has values for minimum and maximum levels of the building (min_level & max_level). Complete SIT tagging is considered any building that has minimum tagging (min_level & max_level) as well as any one of the 4 other elements in the SIT tagging schema [1].

IV. Results & Conclusion

Using Overpass Turbo, data was collected and is displayed in Table 1. Overall, there is a sharp contrast to the amount of data available for buildings, compared to indoor data. OSM does not have enough indoor map data for widespread practical uses. Considering indoor maps are at the early stages of existence, this is not surprising. There is a massive amount of data needed to produce the level of detail needed for wayfinding applications to reach their full potential.

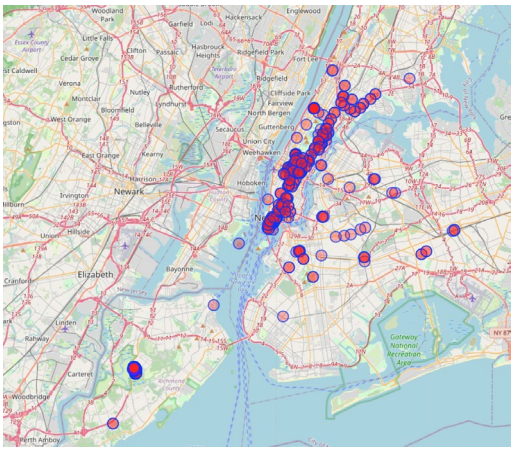
Additionally, it can be observed in Table 1 that there is a discrepancy in the usage of SIT tags and the "indoor=yes" tag. This demonstrates the lack of synchronization amongst the OSM crowdsourcing contributors, showing the flexibility of OSM has both its advantages and disadvantages.

TABLE I
OpenStreetMap Data demonstrating the use of
Simple Indoor Tagging in several cities in USA

City	New York	San Fran.	Chicago	Denver	D.C.	Wichita
Elements with "building" tag	1081795	160377	816930	194855	161107	18956
Elements with "indoor = yes" tag	1079	383	389	227	12	0
Elements with any SIT-Tag	607	230	1441	3755	377	32
Buildings (SIT minimal)	21	10	1	1	0	0
Buildings (SIT complete)	2	8	1	1	0	0

Although OSM has a long way to go until widespread use of indoor navigation is feasible, SIT tagging has shown promising growth. Figure 5 shows the SIT tags that are available in New York City. This shows that crowdsourcing indoor data has practical value, and that OSM contributors are capable of sustained contribution. Crowdsourcing has the potential to make SIT tagging a powerful means to advance wayfinding applications, and OSM has proven that crowdsourcing is a viable option to generate indoor map data.

It is important that researchers investigate the needs of indoor mapping, evaluate the best approach to creating a data model, and predict the best method of collecting data.



[Figure 5. OpenStreetMap Simple Indoor Tagging data elements in New York, New York, USA http://overpass-turbo.eu/ *Accessed Aug. 08, 2020*]

V. Future Work

As OSM is entirely crowdsourced, the future of indoor mapping in OSM depends on a collaborated effort. There are many ways to contribute to the map [17]. OsmInEdit is a notable indoor editor that is based on SIT Tagging [16]. Future work could target the collection of data as well as the methods used to collect data.

Indoor mapping can utilize the existing floor plans of buildings [22]. There is no data centric format of floor plans that allows a simple translation to a geospatial data model. Computer vision is being utilized to collect and process data to overcome this. OsmInEdit includes the option to import a floor plan to map indoor areas in OSM.

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